Humanoid Football-Kicking Robot

Embedded Systems Course

📌 Overview

Design and build a **low-cost** miniature humanoid robot that detects a football using a camera, computes leg joint angles through forward and inverse kinematics, and executes a kicking motion with servo control.

PART 1 — Mechanical Design

Design a low-cost, simplified bipedal lower body with two leg segments per kicking leg, each driven by a servo motor. The robot must stand stably on a flat surface using a fixed base.

PART 2 — Robot Kinematics

Forward Kinematics (FK) answers the question: given joint angles θ_1 and θ_2, where is the foot? Deriving FK is the essential foundation; inverse kinematics simply inverts this model. FK is also used at runtime to validate IK output and catch errors.

2-DOF Planar Leg — FK Equations

$$x = L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1 + \theta_2)$$

$$y = L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1 + \theta_2)$$

Where: L_1 = thigh length, L_2 = shin length, θ_1 = hip angle, θ_2 = knee angle relative to thigh

Inverse Kinematics (IK) Given a desired foot target position (x, y) received from the vision system, the IK solver computes the required hip angle θ_1 and knee angle θ_2. These are then mapped to servo PWM values, and the actuators are commanded accordingly.

• Use the law of cosines to solve for joint angles from link lengths L_1, L_2, and target position

• Map computed angles to PWM values and command servos in real time

PART 3 — Computer Vision: Ball Detection

Use a camera module (ESP32-CAM, USB webcam) to detect the position of a football in real time and communicate the result to the microcontroller.

Send detected ball coordinates from the vision system to the microcontroller. The MCU uses these coordinates as the IK target position.

PART 4 — Report Structure

The written report is a key deliverable. Follow this exact structure:

Chapter 1: Introduction

- Historical background
- Problem statement
- Aims and objectives of the project
- Report agenda

Chapter 2: Literature Review

A thorough survey of prior work. Cover:

- Design & mechanical
- Electronics & software
- Problems & limitations of prior work
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Chapter 3: Proposed System

- System overview
- Improvements over existing humanoid approaches
- System Block Diagrams (Hardware and Software)

Chapter 4: Implementation

- Mechanical design
- State Machines
- Implemntation
- Bill of Materials (BoM)

Chapter 5: Results

- FK accuracy: measured vs. computed foot position
- IK accuracy: target vs. achieved angle
- Vision latency measurement and detection accuracy

Chapter 6: Conclusion

Challenges (NOT Required, but if you did, it would be a bonus)

1. MATLAB / Simulink simulation

Before or alongside hardware implementation, build a MATLAB simulation of the 2-DOF leg kinematics. Use Simulink to model the servo dynamics.

Recommended resource: Humanoid Robot MATLAB tutorial.

2. Wireless dashboard

Build a simple web dashboard (served from an ESP32 or a companion device) that visualizes live telemetry: ball coordinates, joint angles, and state-machine status. Display using a browser over Wi-Fi.

Adding any other enhancement will be considered for bonus marks.

Full Deliverables

Students must submit a complete project package that includes the following:

1. Source Code
2. Report (PDF)
3. Presentation
4. Marketing poster
5. Demo video
6. Any Additional Materials (Simulations, Dashboard, Screenshots)

Academic Integrity & Resources

The use of Large Language Models (LLMs) such as ChatGPT, Claude, and Gemini to generate code for this project is **strictly prohibited**. Submitting code generated by an LLM will be flagged for similarity with other submissions, as LLMs often produce similar solutions, which will result in a penalty for plagiarism. Students are permitted to use LLMs only to learn new concepts or search for resources.